

[A-28]

Q-1 Select the correct option from the following questions.**10**

- 1 _____ is way to simplify the equation.
A. Boolean Algebra B. K-MAP C. BOTH D. NONE
- 2 $A + A'B + A'B' = ?$
A. 1 B. 0 C. A D. A'
- 3 In k-map, octets eliminates _____ variable.
A. one B. two C. three D. four
- 4 The relationship between a function and its binary variables can be represented in _____.
A. truth table B. decoder C. encoder D. multiplexer
- 5 A combinational circuit that performs the arithmetic addition of two bits is called _____.
(A) Full Adder (B) Half Adder (C) Binary Adder (D) Decoder
- 6 A _____ is a memory element that stores a binary digit.
A. binary adder B. decoder C. multiplexer D. flip-flop
- 7 In D flip-flop, when CLK is low then input is _____.
A. high B. low C. Don't care D. Not change
- 8 A multiplexer also called a _____.
A. data multiplier B. data selector C. data inverter D. data remover
- 9 The NOR gate has two or more input signals. If all inputs are _____, the output is high.
A. low B. high C. both A and B D. none
- 10 7. In Comparator, _____ gate is use for comparing bits in word.
A XOR B AND C NOR D XNOR

Q-2 Answer the following questions. (ATTEMPT ANY TEN)**20**

- 1 Describe the NAND, NOR gate.
- 2 Explain commutative law.
- 3 Draw the circuit for : $(A \oplus B)'(BC)(A \oplus C)$
- 4 Describe quad in k-map

(P.T.O.)

- 5 Simplify using k-map $F(A,B,C)=\Sigma(1,2,5)$
- 6 Describe encoder in short.
- 7 Draw the circuit of half adder.
- 8 Describe binary adder in short.
- 9 Define : Flipflop, Register
- 10 Explain D flip-flop in short.
- 11 Define : shift register. What are the types of shift register?
- 12 Draw the circuit of Ring counter.

Q-3 A Write note on: De'Morgan's first and second theorems. 10

OR

Q-3 B Explain AND, XOR, NAND gates. 10

Q-4 A Explain 3x8 line decoder in detail. 5

B What is k-map? Explain octet with example. 5

OR

Q-4 A Explain SOP in detail. 5

B Explain comparator with example. 5

Q-5 A Explain full adder in detail. 4

B Explain 4x1 multiplexer 6

OR

Q-5 A Explain Half adder with an example. 4

B Explain binary subtractor in detail. 6

Q-6 A Explain shift left register. 5

B Explain ring counter in short. 5

OR

Q-6 A Explain shift-right register. 5

B Explain controlled buffer register. 5

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